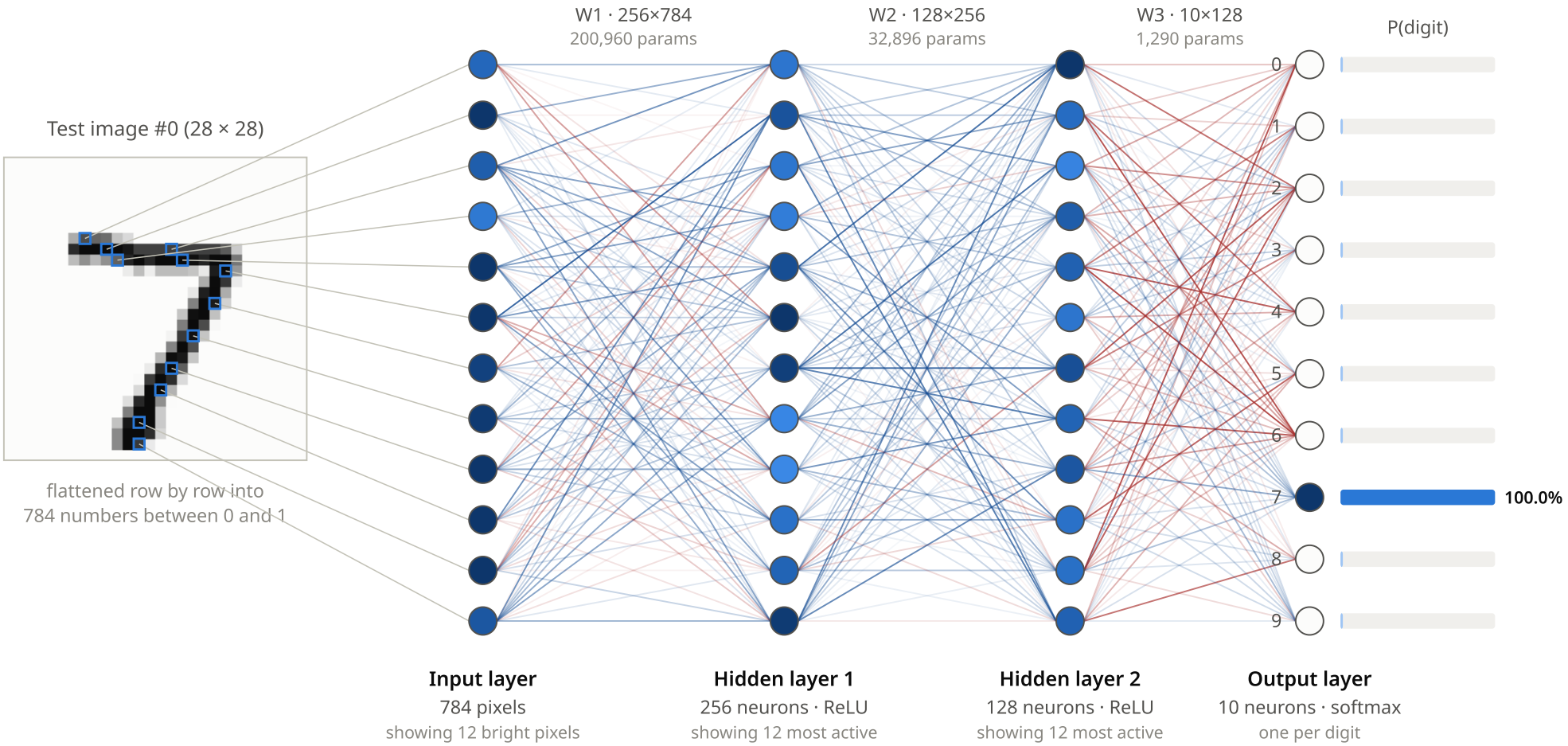
How the network reads a handwritten digit

A fully connected network turns 784 pixel brightnesses into 10 probabilities. Each layer multiplies the previous layer by a weight matrix, adds a bias, and applies an activation. Below, real test image #0 (a 7) flows through the trained model. It predicts **7** with 100.00% confidence.



positive weight · negative weight · line opacity = |weight| · strongly active · quiet

235,146 trainable parameters · 98.17% test accuracy

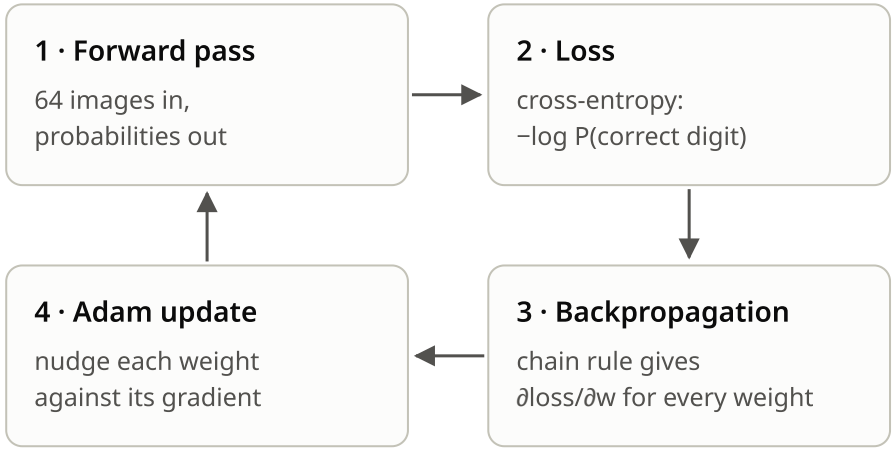
Every layer is a matrix multiply plus a simple non-linearity

Hidden layers use ReLU (max(0, z)), so a neuron either stays silent or passes its signal on. For any given digit, only part of the network lights up. The output layer uses softmax to turn 10 scores into probabilities that sum to 1.

The math, with shapes

x	pixels / 255	784
$h_1 = \mathrm{ReLU}(W_1 x + b_1)$	first hidden layer	256
$h_2 = \mathrm{ReLU}(W_2 h_1 + b_2)$	second hidden layer	128
$p = \mathrm{softmax}(W_3 h_2 + b_3)$	$p_i = e^{z_i} / \Sigma\, e^{z_j}$	10
$\mathrm{prediction} = \mathrm{argmax}(p)$	most likely digit	1

How it learned the weights

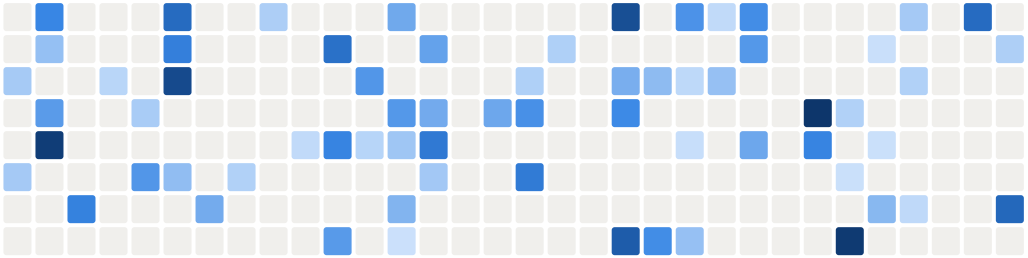


860 batches per epoch × 15 epochs ≈ 12,900 updates

During training, 20% of hidden neurons are randomly switched off on each step (dropout). This stops the network from relying on any single neuron.

Hidden activity for image #0

Hidden layer 1: **66** of 256 neurons fire



Hidden layer 2: **51** of 128 neurons fire



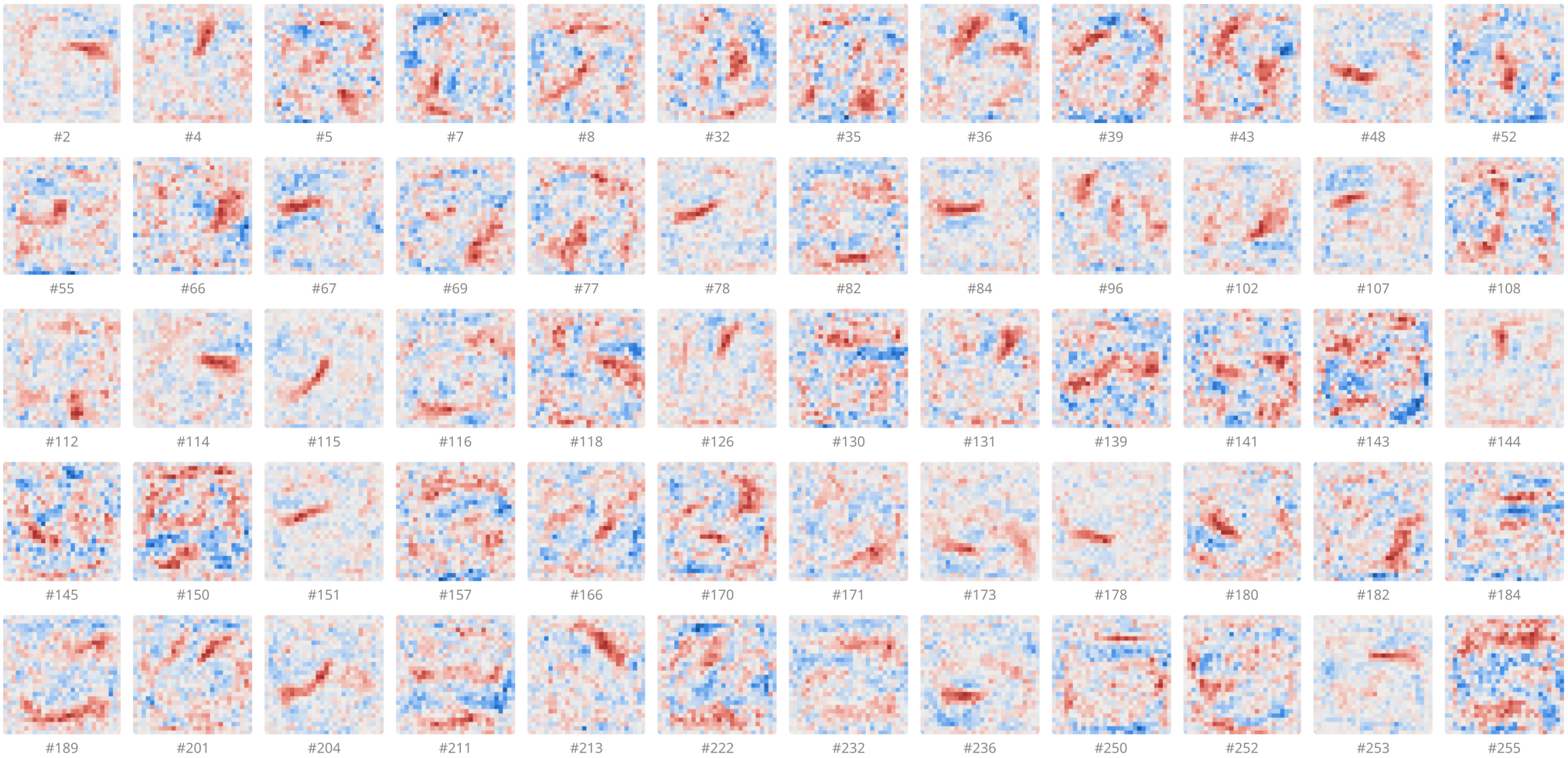
silent (ReLU output 0) weak strong

Each square is one neuron, in index order. A different digit lights up a different pattern. The output layer reads these patterns to decide which digit it sees.

Shapes are per image. Training runs 64 images at a time.

What the first layer learned to look for

Each neuron in hidden layer 1 has 784 weights, one per pixel, so its weights can be drawn as a 28×28 image. Blue pixels push the neuron to fire when ink is there. Red pixels hold it back. Together they act as stroke and edge detectors. These are the 60 neurons with the largest weights.



negative weight (inhibits) 0 positive (excites)

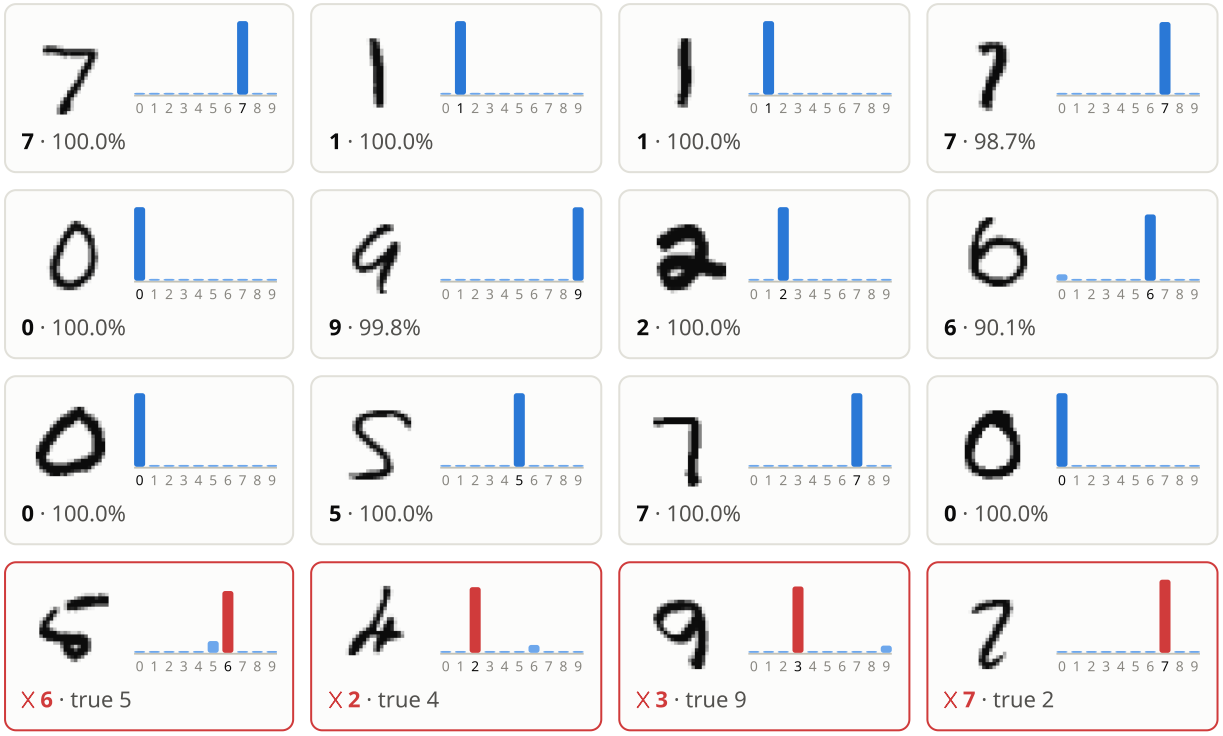
Each map is scaled to its own largest |weight|.

RESULTS ON 10,000 UNSEEN TEST IMAGES

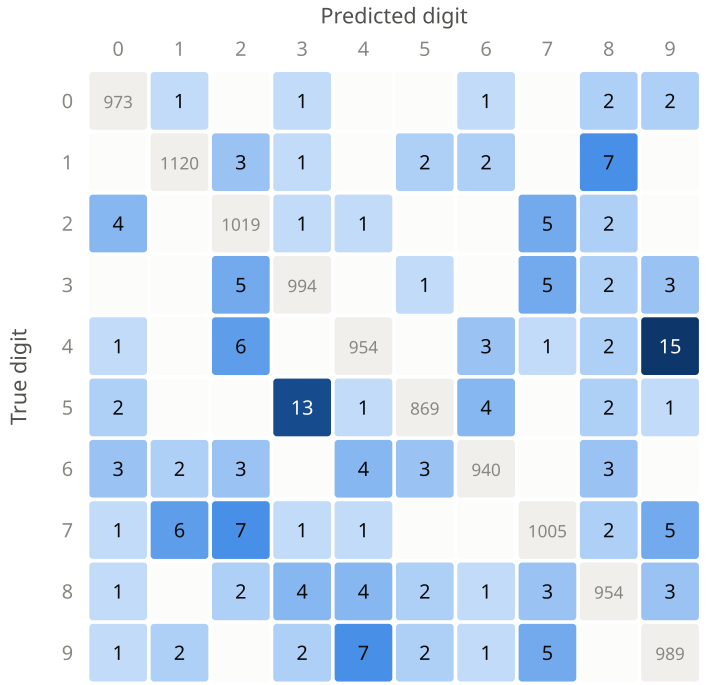
Predictions, and where it gets confused

The network never saw these images during training. It gets 9,817 of 10,000 right (98.17%). Most mistakes are on digits that are genuinely ambiguous or badly written.

Sample predictions: 12 correct, then the first 4 mistakes



Confusion matrix: off-diagonal cells are errors



Diagonal cells (gray) count correct answers. Darker blue means more errors. For example, the cell at row 4, column 9 shows that 15 images of a 4 were read as a 9.

Bars show the softmax probability for each digit. Red marks a wrong prediction.

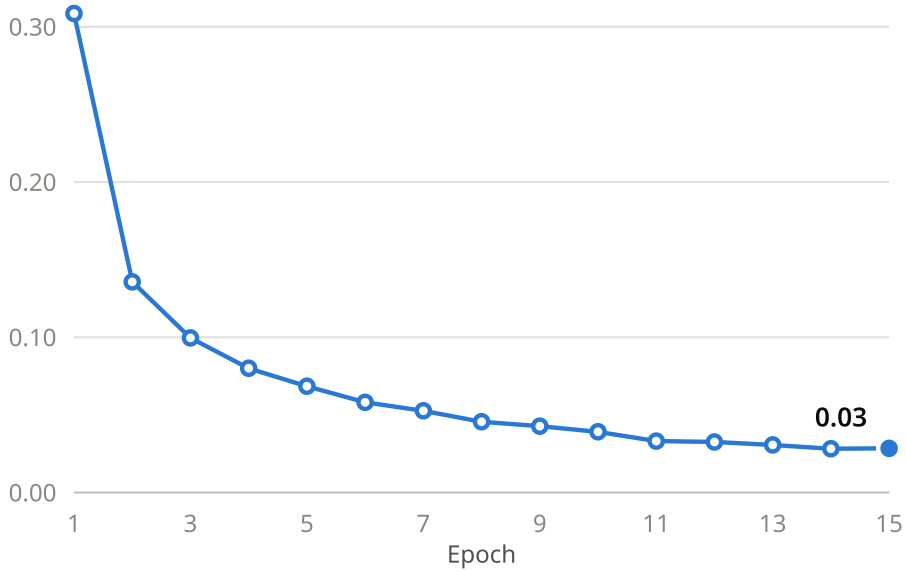
TRAINING RUN

How it improved, epoch by epoch

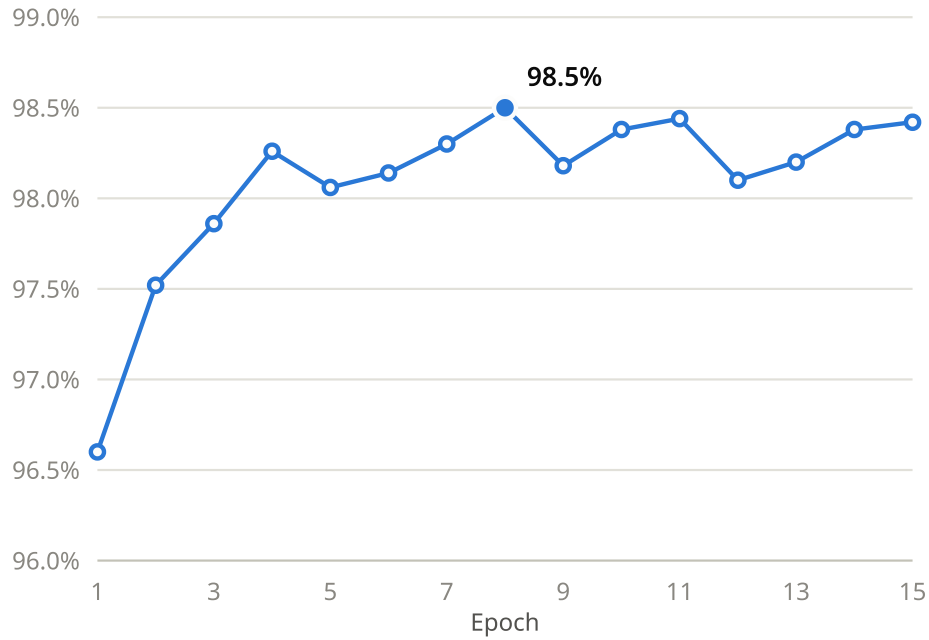
One epoch is one pass over the 55,000 training images. Loss keeps falling, but validation accuracy levels off after a few epochs. The model saved to disk is the one with the best validation accuracy (epoch 8).

98.17%	98.50%	235,146	15
test accuracy	best validation accuracy (epoch 8)	trainable parameters	epochs · Adam, lr 0.001, batch 64

Training loss (cross-entropy)



Validation accuracy (axis starts at 96%)



Validation set: 5,000 images held out from the training data.